

Machine Learning Fundamentals

Chapter 1: What is Machine Learning?

Machine learning is a subset of artificial intelligence that focuses on building systems that learn from data. Instead of being explicitly programmed, these systems improve their performance through experience.

There are three main types of machine learning:

1. Supervised Learning - learns from labeled data
2. Unsupervised Learning - finds patterns in unlabeled data
3. Reinforcement Learning - learns through trial and error

Chapter 2: Key Concepts

Feature: An individual measurable property of the data being observed.

Label: The output variable that we are trying to predict.

Training: The process of teaching the model using historical data.

Overfitting: When a model learns the training data too well and performs poorly on new, unseen data.

Chapter 3: Common Algorithms

Linear Regression: Used for predicting continuous values.

Decision Tree: A tree-like model of decisions.

Neural Network: Inspired by biological neural networks in the brain.

Random Forest: An ensemble method that combines multiple decision trees.

Chapter 4: Model Evaluation

To evaluate how well a model performs, we use metrics such as:

- Accuracy: The ratio of correct predictions to total predictions
- Precision: The ratio of true positives to all predicted positives
- Recall: The ratio of true positives to all actual positives
- F1 Score: The harmonic mean of precision and recall